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Project Specifications

	Assigned Spec.	Modifications	Rationale
Platform	CubeSat 6U	–	–
Attitude parameters	Direction Cosines	–	–
Mandatory sensor	Sun Sensor	+ 1 Magnetometer + 1 Gyroscope	Attitude Determination Eclipse Propagation
Actuators	3 Reaction Wheels	–	–

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Table 1: Nomenclature

Symbol	Meaning	Symbol	Meaning
$\vec{\omega}$	Angular velocity (body frame)	$\alpha_{\oplus}$	Apparent angular radius of Earth
$\mathbf{A}_{B/N}$	Direction Cosine Matrix (Body→Inertial)	ν	Shadow factor (0=umbra, 1=sunlight)
$\mathbf{A}_{B/N}^U$	Estimated DCM (sensor output)	ε_i	Angular sensor error
$\mathbf{A}_{B/L}$	Target DCM (Body→LVLH)	σ	Noise standard deviation
$\vec{T}_{ctrl}$	Control torque (reaction wheels)	N_d	Noise spectral density
$\vec{T}_{gg}$	Gravity gradient torque	T_s	Sampling period
$\vec{T}_{mag}$	Magnetic torque	$\mathbf{S}_{nl}$	Gyroscope non-linearity
$\vec{b}_B$	Magnetic field in body frame	$\vec{b}_{bias}$	Sensor bias
$\vec{s}_B$	Sun direction in body frame	$\vec{\eta}_{tot}$	Total noise (white+pink+brown)
$\vec{h}$	Reaction wheel angular momentum	$\vec{u}_{ctrl}$	Control signal (controller output)
$\mathbf{I}_b$	Inertia tensor	ARW	Angle Random Walk (gyro)
K_{SF}	Scale factor error (gyro)	ZOH	Zero-Order Hold
K_{nl}	Non-linearity coefficient (gyro)	FOV	Field of View
$\mathbf{C}_{mis}$	Sensor misalignment matrix	DCM	Direction Cosine Matrix
$\mathbf{C}_{non}$	Sensor non-orthogonality matrix	ADCS	Attitude Determination and Control System
$\alpha_{\odot}$	Apparent angular radius of Sun	RW	Reaction Wheel

1 Introduction

The primary objective of this study is the design of the Attitude Determination and Control System (ADCS) for a 6U CubeSat dedicated to *Environmental Monitoring*. To ensure effective remote sensing operations, the ADCS is required to guarantee a pointing accuracy of 1° towards the Earth (nadir pointing). Tab. 2 defines the selected parameters.

Table 2: Orbital Parameters and Environmental Rationale

Parameter	Value	Scientific & Engineering Details
Semi-major axis	$a = 9371$ km	Periapsis radius $r_p \approx 7502$ km ($h_p \approx 1124$ km). Apoapsis radius $r_a \approx 11254$ km ($h_a \approx 4876$ km).
Eccentricity	$e = 0.2$	Allows to perform vertical profiling of the Earth's atmosphere.
Inclination	$i = 30^\circ$	Covers equatorial and mid-latitude regions, for environmental monitoring.

This orbital regime dictates a specific hierarchy of environmental disturbances, distinct from standard CubeSats' LEOs for remote sensing, as detailed in Sec. 2.4.2.

PHYSICAL WORLD

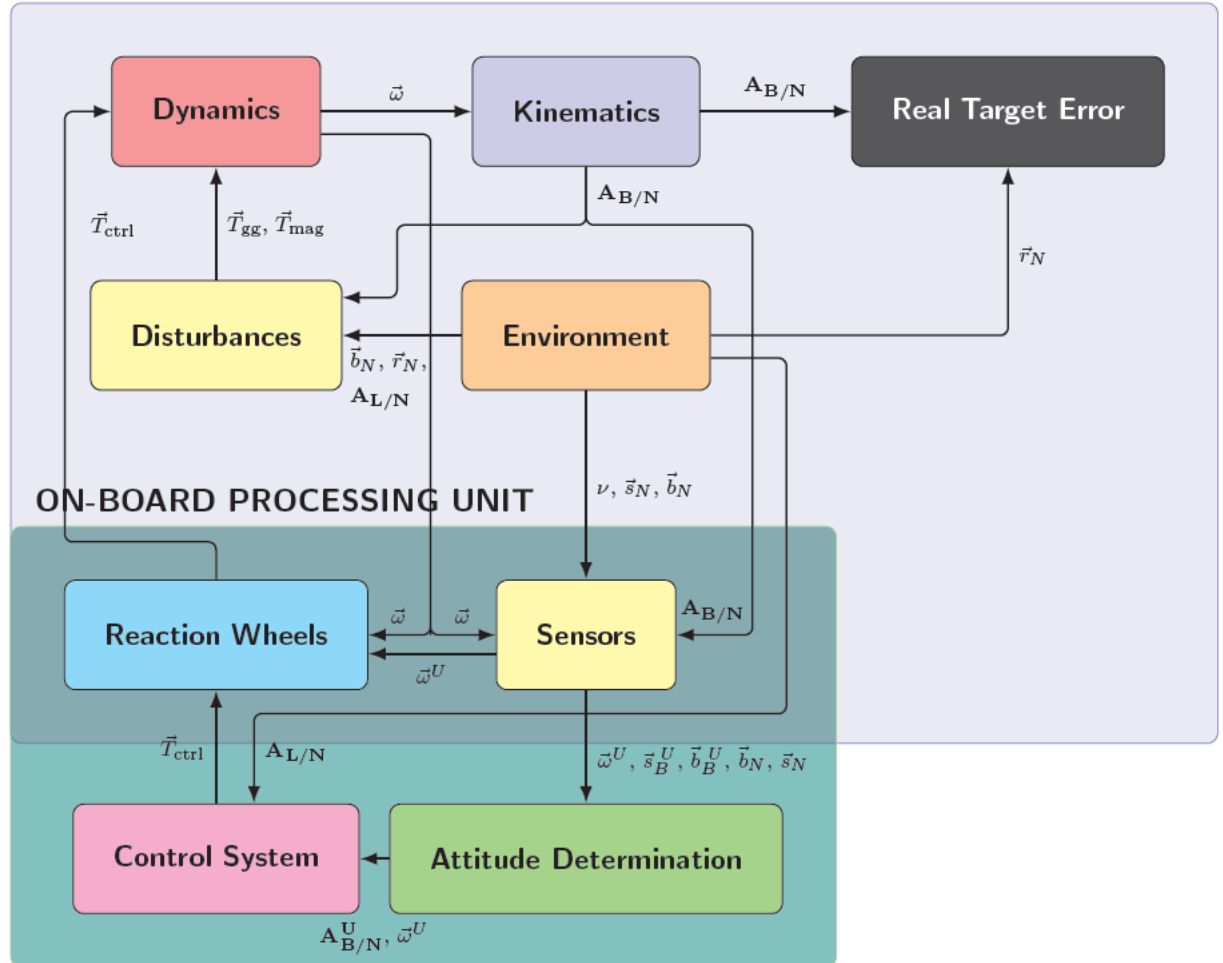


Figure 1: Architecture of the Attitude Determination and Control System (ADCS).

2 Environment and Spacecraft Motion

2.1 Spacecraft Mass Properties and Inertia Tensor

The spacecraft structure follows the 6U CubeSat standard form factor with envelope dimensions $L = 30$ cm, $W = 20$ cm, and $H = 10$ cm. ^[11] For this mission analysis a total mass of $m_b = 12$ kg is assumed, representing a standard configuration. ^[8]

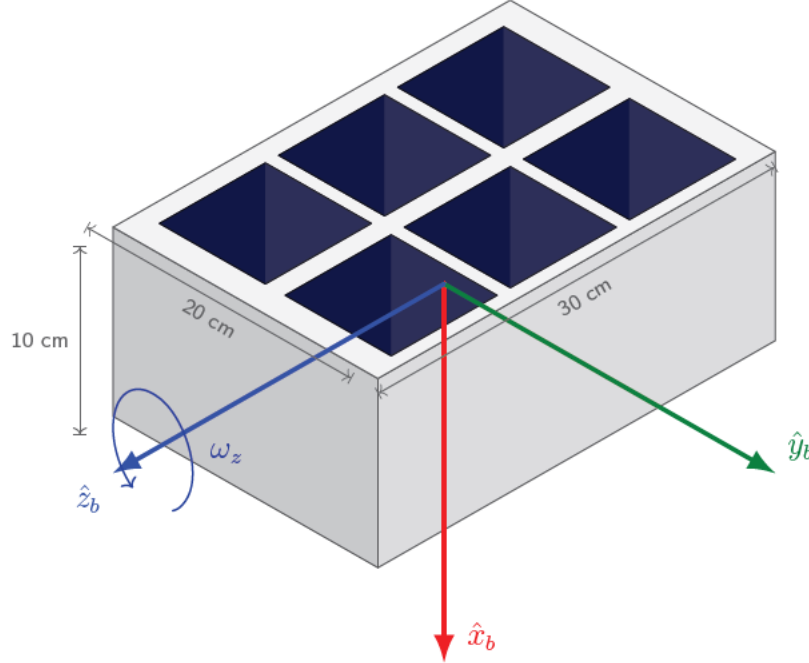


Figure 2: 6U CubeSat configuration ($30 \times 20 \times 10$ cm, 12 kg). The Body Reference Frame is aligned with the principal axes: $\hat{x}_b$ (max inertia) corresponds to the thickness axis, while $\hat{z}_b$ (min inertia) corresponds to the longitudinal axis, which is the nominal rotation axis.

The inertia tensor is derived analytically by modeling the spacecraft as a simple cuboid rotating about its center of mass which is positioned as its center of geometry. Assuming the solar panels are body-mounted on the Zenith-facing surface ($S_{\text{solar}} \approx 30 \times 20$ cm), the inertia tensor $\mathbf{I}_b$ in the body frame is defined as follows:

$$\mathbf{I}_b = \begin{bmatrix} 0.1300 & 0 & 0 \\ 0 & 0.1000 & 0 \\ 0 & 0 & 0.0500 \end{bmatrix} \text{ kg} \cdot \text{m}^2 \quad (1)$$

2.2 Spacecraft Dynamics

The dynamics module is responsible for solving the rotational equations of motion of the spacecraft. Consistent with the rigid body assumption and the inertia tensor provided in the previous section, the model implements Euler's equations for rotational dynamics.

The system computes the angular acceleration $\dot{\vec{\omega}}$ in the body frame based on the total moment applied to the body, which is the sum of external environmental disturbances and the control torque provided by the reaction wheels. The governing equation is: ^[8]

$$\mathbf{I}_b \dot{\vec{\omega}} + \vec{\omega} \times (\mathbf{I}_b \vec{\omega}) = \vec{T}_{\text{tot}} = \vec{T}_{\text{gg}} + \vec{T}_{\text{mag}} + \vec{T}_{\text{ctrl}} \quad (2)$$

The block integrates $\dot{\vec{\omega}}$ to output the angular velocity state vector $\vec{\omega} = [\omega_x, \omega_y, \omega_z]^T$, which is then fed into the *Kinematics* block and the sensor models. The coupling term $\vec{\omega} \times (\mathbf{I}_b \vec{\omega})$ accounts for the gyroscopic stiffening effects inherent to the 3D rotation.

2.3 Spacecraft Kinematics

The kinematics module propagates the attitude of the spacecraft relative to the Inertial Frame (ECI). In compliance with the project requirements the attitude is represented using the *Direction Cosine Matrix* (DCM), denoted as $\mathbf{A}_{\mathbf{B}/\mathbf{N}}$.

The kinematic propagation is implemented by integrating Poisson's kinematic differential equation. Given the skew-symmetric matrix $[\boldsymbol{\omega}\wedge]$ associated with $\vec{\omega}$, the rate of change of the attitude matrix is computed as:

$$\dot{\mathbf{A}} = -[\boldsymbol{\omega}\wedge]\mathbf{A}, \quad [\boldsymbol{\omega}\wedge] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad (3)$$

The solver integrates $\dot{\mathbf{A}}$ to obtain the instantaneous attitude matrix $\mathbf{A}_{\mathbf{B}/\mathbf{N}}(t)$. To counteract numerical drift inherent to the integration process, a orthogonalization is performed at each simulation step to ensure that $\mathbf{A}_{\mathbf{B}/\mathbf{N}}$ remains strictly orthogonal. This was validated also performing the integration of Dynamics through *quaternions*, which were later transformed into DCM, obtaining the same results.

2.4 Environment & Disturbances

In the *Environment* block the orbit designed in the Sec. 1 is propagated, the CubeSat's target attitude matrix $\mathbf{A}_{\mathbf{B}/\mathbf{L}}$ is evaluated and the eclipse and the magnetic field is modeled. The *Disturbances* block computes the torque affecting the spacecraft.

2.4.1 Eclipse Propagation and Shadow Model

The implemented eclipse model adopts a *dual-cone* approach, which accounts for the finite angular dimensions of both the Sun and the Earth. Unlike simple cylindrical shadow models — which treat the light source as a point — this method resolves the *umbra* (total eclipse) and *penumbra* (partial eclipse) regions, ensuring a smooth transition of the solar flux in the shadow factor $\nu \in [0, 1]$. Furthermore, the Earth is modeled as an oblate ellipsoid, ensuring that the occulting body's apparent radius varies correctly with the satellite's latitude. ^[10]

Given the Earth's equatorial radius R_{eq} and polar radius R_{pol} , and defining the satellite's geocentric position vector as $\mathbf{r}_{\text{sat}} = [x, y, z]^T$, the geocentric latitude ϕ is derived:

$$\sin \phi = \frac{z}{\|\mathbf{r}_{\text{sat}}\|}, \quad \cos \phi = \sqrt{1 - \sin^2 \phi} \quad (4)$$

The local Earth radius $R_{\oplus, \text{loc}}$ projecting the shadow is calculated using an ellipse geometry, implying that the shadow cone is wider when the satellite overflies the equator:

$$R_{\oplus, \text{loc}} = \sqrt{\frac{(R_{\text{eq}}^2 \cos \phi)^2 + (R_{\text{pol}}^2 \sin \phi)^2}{(R_{\text{eq}} \cos \phi)^2 + (R_{\text{pol}} \sin \phi)^2}} \quad (5)$$

Let $\vec{r}_{\text{sat}}$ and $\vec{r}_{\odot}$ denote the ECI position vectors of the satellite and the Sun. From the apparent angular radii of the Sun $\alpha_{\odot}$ and the Earth $\alpha_{\oplus}$ from the spacecraft, the angular separation ζ between the center of the Sun and the center of the Earth is computed using the dot product of their direction unit vectors. The *visibility* (or *shadow*) *factor* ν is defined as the ratio of the unblocked to the total apparent solar area ($A_{\odot} = \pi\alpha_{\odot}^2$).

The algorithm distinguishes three geometric cases:

1. **Full Sunlight** ($\nu = 1$): The angular separation is large enough that the solar and terrestrial disks do not overlap, and $\zeta \geq (\alpha_{\odot} + \alpha_{\oplus})$.
2. **Umbra** ($\nu = 0$): The Earth's disk completely covers the Sun ($\zeta \leq (\alpha_{\oplus} - \alpha_{\odot})$).
3. **Penumbra** ($0 < \nu < 1$): The two disks intersect. The overlapping area A_{int} is calculated from the geometry of circular segment intersection. From the radial distance x from the center of the Sun to the chord of intersection and the half-length of the chord y , the intersection area is obtained and therefore the shadow factor ν :

$$x = \frac{\alpha_{\odot}^2 - \alpha_{\oplus}^2 + \zeta^2}{2\zeta}, \quad y = \sqrt{\alpha_{\odot}^2 - x^2}$$

$$A_{\text{int}} = \underbrace{\alpha_{\odot}^2 \arccos\left(\frac{x}{\alpha_{\odot}}\right) + \alpha_{\oplus}^2 \arccos\left(\frac{\zeta - x}{\alpha_{\oplus}}\right)}_{\text{Sum of the two circular sectors}} - \underbrace{\zeta y}_{\text{Central kite area}}$$

$$\Rightarrow \boxed{\nu = 1 - \frac{A_{\text{int}}}{\pi\alpha_{\odot}^2}}$$

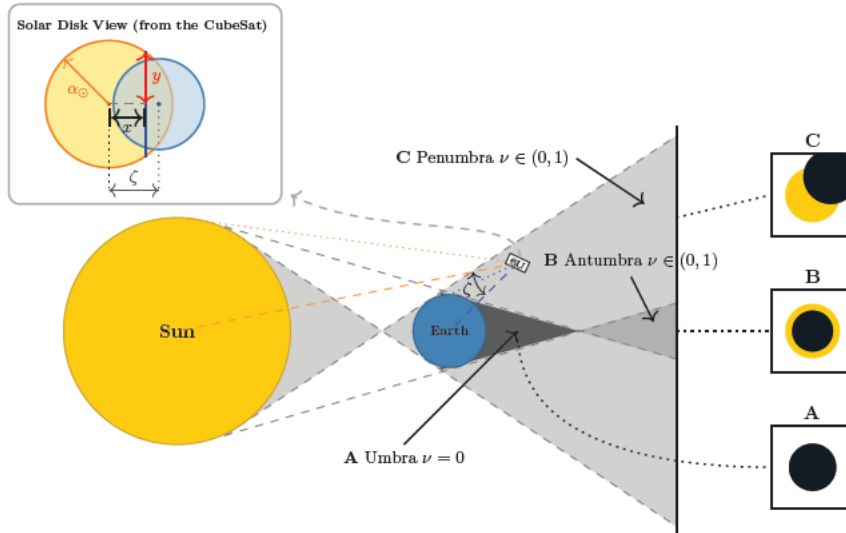


Figure 3: Schematic representation of the shadow zones generated by the Earth (not to scale).

2.4.2 Disturbances

The preliminary design includes an order-of-magnitude analysis of the environmental disturbance torques acting on the spacecraft attitude, which are summarized in Tab. 3.

The reference torques were computed using the following assumptions:

- The magnetic dipole moment, consistent with typical 6U CubeSat specifications, is:

$$\vec{j}_B = [0.01 \quad 0.05 \quad 0.01]^T \text{ A} \cdot \text{m}^2 \quad \Rightarrow \quad \vec{M} = \vec{j}_B \times \vec{b}_B \quad (6)$$

- Gravity gradient torque is derived from the spacecraft's principal moments of inertia. Given the mean angular velocity n along the orbit, the preliminary torque is computed:

$$M = 3n^2 (I_{\max} - I_{\min}) \quad (7)$$

- Solar Radiation Pressure (SRP) and atmospheric drag assume a Center of Gravity (CoG) offset with respect to the Center of Pressure (CoP) of 0.01 m due to the symmetrical geometry. Drag is evaluated at pericenter density ($\rho = 6.97 \times 10^{-17} \text{ kg/m}^3$) and velocity, assuming a flat-plate drag coefficient $C_D = 2.2$ and maximum cross-sectional area $A = 0.03 \text{ m}^2$.

Table 3: Magnitude of Disturbances

Source of disturbance	Torque [Nm]
Magnetic Torque	3.3676×10^{-7}
Gravity Gradient	5.8125×10^{-8}
Solar Radiation Pressure	4.7433×10^{-9}
Air Drag	2.10555×10^{-13}

Gravity gradient and magnetic torque are chosen, being the most notable disturbances.

Magnetic Field and Magnetic Torque The geomagnetic field is modeled using the International Geomagnetic Reference Field (IGRF), developed by the International Association of Geomagnetism and Aeronomy (IAGA). The recursive formulation of the model has been directly implemented within the Simulink[®] environment [1]. Based on the spacecraft position, the selected model degree (up to 13), and the IGRF coefficients, the implementation returns the magnetic field vector expressed in the inertial reference frame. The IGRF coefficients correspond to the 2025 epoch and were taken from [5].

As a validation method, the results obtained from the implemented model have been compared with the official IGRF magnetic field maps [6]. Fig. 4 shows the comparison between the total magnetic field intensity maps.

The same comparison procedure was applied to the vertical and horizontal components of the magnetic field, as well as to the inclination and declination angles, leading to similarly good results.

The magnetic torque $\vec{M}$ acting on the spacecraft can then be obtained through the vector cross product between the spacecraft magnetic dipole moment and the magnetic field vector $\vec{b}_B$, expressed in the spacecraft body reference frame (Eq. 6)

Gravity Gradient The gravity gradient is a disturbance torque $\propto 1/r^3$ due to the central nature of gravitational attraction that makes the satellite subjected to slightly different forces in different parts of the spacecraft, depending on its attitude.

The model was validated by checking the periodicity of the disturbance with respect to the orbit and all the stability/instability regions of the Kr-Ky diagram. Given the spacecraft mass and geometry, the gravity gradient coefficients are:

$$Kr = -0.8 \quad Ky = -0.3846 \quad (8)$$

These values place the CubeSat in an unstable region in the Kr-Ky stability diagram.

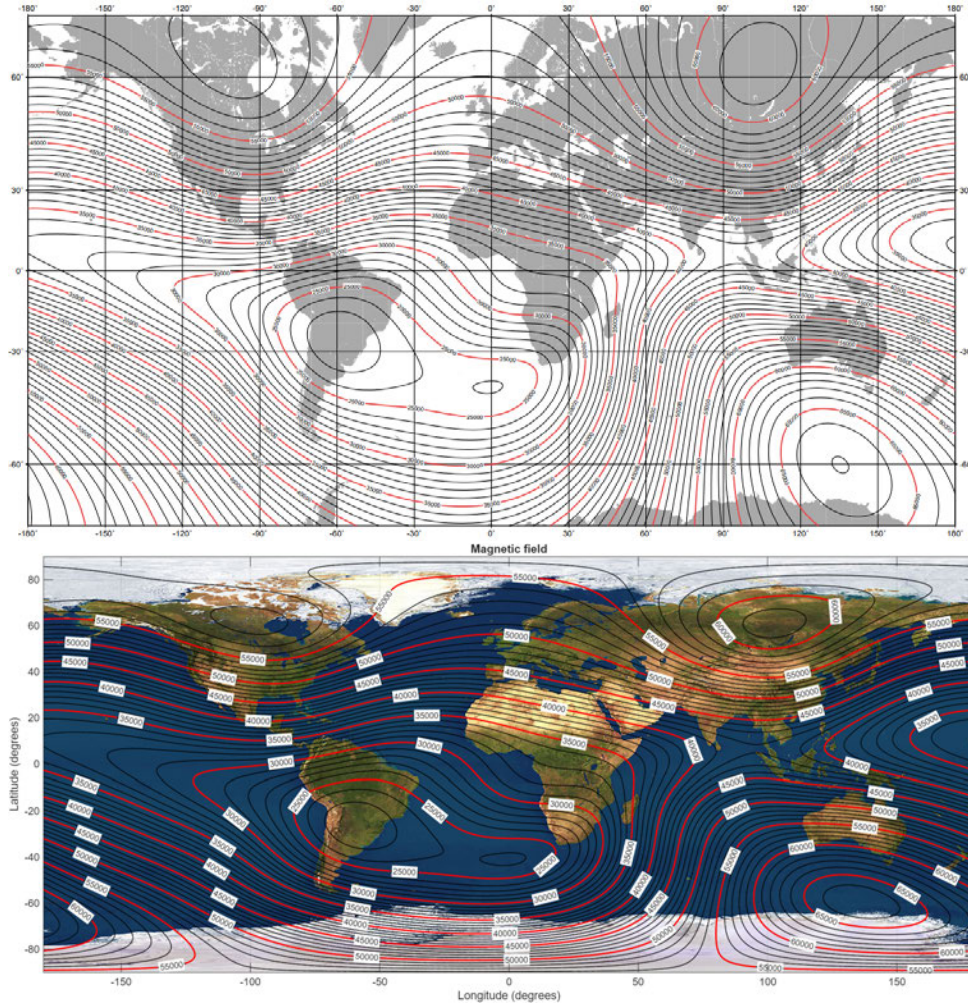


Figure 4: IAGA reference map and implemented map for the magnetic field intensity (nT) ^[1].

3 Sensors & Actuators

Onboard a spacecraft, ADCS sensors are designed to measure specific physical quantities to estimate the spacecraft attitude and angular velocity. Actuators, on the other hand, convert the control torques commanded by the ADCS into actual ones on the S/C itself. Therefore, they are the interface between the processing unit and the spacecraft dynamics.

3.1 Sensors

Standard attitude determination requires at least two non-collinear reference vectors to resolve the spacecraft orientation. Consequently, the preliminary design relied on the mandatory sun sensor and on a magnetometer only. However, due to the presence of the Sun sensor, the spacecraft is unable to determine its attitude during eclipse periods and to perform attitude control.

To overcome this limitation, attitude propagation based on angular velocity measurements is required to estimate the attitude during eclipse. At the beginning, the angular velocity vector was estimated from variations in the magnetic field measured by the magnetometer. Nevertheless, this approach proved to be highly unreliable, as it will be shown in the Orbit Propagation Chapter (see Sec. 4.2.2). For this reason, a gyroscope was deemed necessary and subsequently integrated into the system.

3.1.1 Oustaloup Approximation

The noise affecting the sensors in the following sections is modeled as white and colored noise, composed of pink and brown components. In order to ideally reproduce pink noise, a white random signal must be filtered using a discrete low-pass filter with a slope of -10 dB/decade. Such a filter cannot be analytically derived using classical integer-order systems, since the required slope corresponds to half of a first-order low-pass filter.

To address this limitation, the Oustaloup approximation is adopted. The following considerations are based on Oprzedkiewicz et al. [7] This method enables the approximation of the behavior of a fractional-order system of the form:

$$G(s) = \frac{1}{s^\alpha}, \quad \alpha \in \mathbb{R} \quad (9)$$

In the studied case, the fractional order is set to $\alpha = 0.5$, which corresponds to the desired -10 dB/dec slope.

The numerical approximation of the fractional operator is expressed as:

$$s^\alpha \cong k_f \prod_{n=1}^N \frac{1 + \frac{s}{\mu_n}}{1 + \frac{s}{\nu_n}} \quad (10)$$

where $\gamma = \left(\frac{\omega_h}{\omega_l}\right)^{\frac{\alpha}{N}}$, $\eta = \left(\frac{\omega_h}{\omega_l}\right)^{\frac{1-\alpha}{N}}$, $\mu_1 = \omega_l \sqrt{\eta}$, then:

$$\begin{aligned} \nu_n &= \mu_n \gamma, \quad n = 1, \dots, N \\ \mu_{n+1} &= \nu_n \eta, \quad n = 1, \dots, N-1 \end{aligned} \quad (11)$$

where ω_h and ω_l define the upper and lower angular frequency bounds over which the target slope is to be approximated, and N represents the number of poles and zeros used in the approximation.

Ideally, pink noise exhibits a constant spectral slope of -10 dB/decade over the entire frequency spectrum. In practice, increasing the frequency range degrades the approximation accuracy. Although a higher number of poles and zeros improves the slope fidelity, excessively large values of N may lead to poor numerical conditioning and increased computational cost.

The referenced study recommends $N = 8$ studying frequencies ranges from $\omega_l = 0.001$ rad/s to $\omega_h = 1000$ rad/s. Using these parameters, the corresponding continuous-time transfer function is obtained (see Fig. 5).

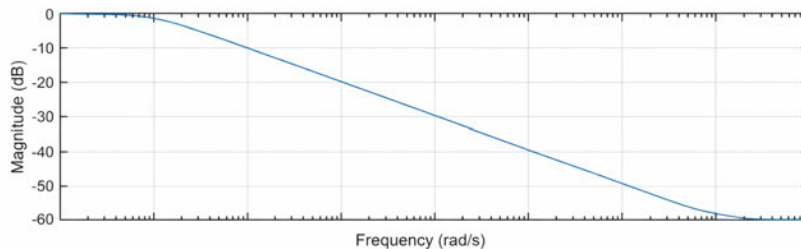


Figure 5: -10 dB/dec Low-Pass Filter.

Finally, the transfer function is converted into a discrete-time representation using the Tustin (bilinear) transformation, enabling its implementation.

3.1.2 Sun Sensor

The Sun sensor provides the direction of the Sun with respect to the spacecraft body frame. For this mission, the selected hardware is the NCSS-SA05 [4]. Based on the sensor datasheet, the parameters relevant for modeling the sensor are summarized in Tab. 4.

First, to introduce latency in the signal evaluation, a *delay* block corresponding to one time step is included.

Table 4: Sun Sensor Performance Characteristics [4]. Shape and mass compatible with 6U format.

Parameter	NCSS-SA05 (Analogue)
Field of View (FOV)	114°
Accuracy (3σ)	< 0.5° RMS
Update Rate	> 10 Hz
Dimensions	33 mm × 11 mm × 6 mm
Mass	< 5 g

The sensor is never perfectly aligned with the spacecraft body axes due to installation tolerances. This misalignment is modeled through a constant rotation matrix around the three body axes $\mathbf{C}_{\text{mis}}$. Assuming small misalignment angles, the rotation matrix is computed using a randomly-generated misalignment with maximum magnitude $\delta = 0.1^\circ$.

The measurement error provided in the datasheet is expressed in angular terms; therefore, sensor noise is modeled as a rotation applied to the Sun direction vector $\vec{s}_B$. The noise is represented as the superposition of three distinct noise components:

$$\eta(f) = \underbrace{N_0}_{\text{White}} + \underbrace{\frac{K_p}{f}}_{\text{Pink}} + \underbrace{\frac{K_b}{f^2}}_{\text{Brown}} \quad (12)$$

where the individual noise components are generated as follows:

- white noise, generated using a *random number generator* block;
- brown noise, obtained by filtering the *random number generator* output with a discrete low-pass filter featuring a -20 dB/dec slope;
- pink noise, obtained by filtering the *random number generator* output with a discrete low-pass filter featuring a -10 dB/dec slope (see Subsec. 3.1.1).

The discrete transfer function of the brown noise, using the bilinear transformation, is:

$$G(z) = \frac{0.001}{\frac{2(z-1)}{Ts(z+1)} + 0.001} \quad (13)$$

The cut-off frequency $\omega = 0.001$ models the majority of the frequencies content with the desired slope.

The root mean square value of the three noises is set to $\sigma = 0.5^\circ$, as reported in Tab. 4, and applied as a gain. This represents a pessimistic assumption, since the superposition of the three noise sources results in a higher overall RMS value. Nevertheless, a conservative scenario was intentionally selected to demonstrate the robustness of the model. Finally, a small random bias error with maximum magnitude $b = 0.1^\circ$ is added. The resulting measured Sun vector is expressed as a function of the time-varying rotation matrix $\mathbf{C}_{\text{noise}}(\eta, b)$ resulting from the constant bias b and random noise η :

$$\vec{s}_B^U = \mathbf{C}_{\text{noise}}(\eta, b) \cdot \mathbf{C}_{\text{mis}}(\delta) \cdot \vec{s}_B \quad (14)$$

Finally, the signal is passed through a *Zero-Order Hold* block to simulate the digital sensor update rate of 10 Hz.

During the eclipse, the output is set to -2 , to be used by the control logic of the attitude determination by an enable block, disabling the subsystem when $\nu = 0$. An example output signal of the sun sensor over two orbital periods is shown in Fig. 6.

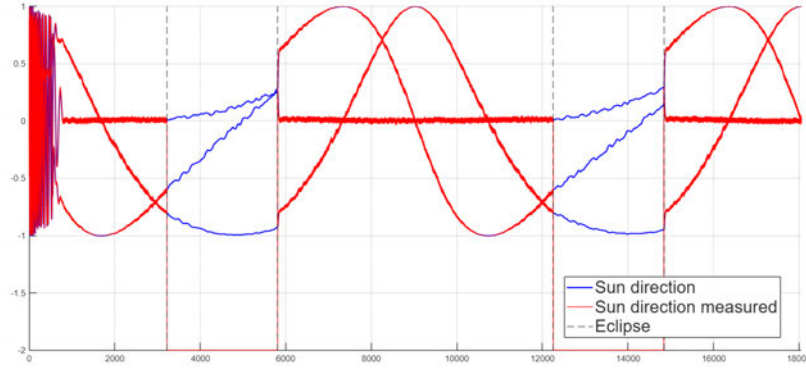


Figure 6: Sun sensor output over two orbits, in eclipse the red signal is set to -2 .

The discontinuity at the end of the eclipse is caused by the control action that removes the errors caused by the orbit propagation. Please note that a full field of view for the sun sensor is hypothesized with an appropriate number of sun sensors.

3.1.3 Magnetometer

The tri-axial magnetometer measure the Earth's magnetic field vector expressed in the body frame ($\vec{b}_B$). The selected hardware is the Pegasus Magnetometer by NewSpace Systems [3], which is based on magneto-resistive sensing technology.

Table 5: Pegasus Magnetometer Performance Characteristics [3]

Parameter	Value
Measurement Range	$\pm 60,000$ nT
Orthogonality Error	$\leq 1^\circ$
Resolution	≤ 8 nT
Update Rate	< 25 Hz
Noise Density	< 16 nT / $\sqrt{\text{Hz}}$
Mass	85 g
Dimensions	99 mm $\times$ 43 mm $\times$ 17 mm

The sensor dimensions and mass are compatible with a 6U CubeSat platform.

The magnetometer measurement model maps the true magnetic field vector $\vec{b}_B$ to the digital output versor $\vec{b}_B^U$. The transformation from the ideal orthogonal frame to the actual sensor frame is performed through two cascaded matrix operations.

First, the vector is misaligned with respect to the spacecraft body axes by small angles δ , using the rotation matrix $\mathbf{C}_{\text{noise}}(\delta)$, similarly to the Sun sensor.

Second, the three sensing axes of the magnetometer are not perfectly orthogonal. This imperfection is modeled by the matrix $\mathbf{C}_{\text{non}}$, which is constructed from three small angles ε_1 , ε_2 , and ε_3 . As implemented in the simulation, this matrix introduces cross-axis coupling effects with a lower triangular matrix:

$$\mathbf{C}_{\text{non}} \approx \begin{bmatrix} 1 & 0 & 0 \\ \varepsilon_1 & 1 & 0 \\ \varepsilon_2 & \varepsilon_3 & 1 \end{bmatrix} \quad (15)$$

The datasheet specifies an orthogonality error $\leq 1^\circ$, which bounds the values of ε_i . The complete measurement equation implemented in Simulink[®] is given by:

$$\vec{b}_B^U = \text{ZOH} \left[\text{Sat} \left(\mathcal{Q} \left(C_{\text{non}} C_{\text{mis}} \vec{b}_B + b_{\text{bias}} + \eta_{\text{total}} \right) \right) \right] \quad (16)$$

where b_{bias} is a small bias vector of arbitrary magnitude 15 nT, η_{total} is the total stochastic noise vector, and $\mathcal{Q}(\cdot)$, $\text{Sat}(\cdot)$, and $\text{ZOH}(\cdot)$ represent quantization ($q = 8$ nT), saturation check ($\pm 60 \mu\text{T}$), and temporal discretization respectively.

Unlike the Sun sensor, in this case bias and noises are not applied as rotational errors, since the datasheet specifications are provided in magnetic field units. Consequently, they are directly added to the magnetic field components after accounting for misalignment and non-orthogonality. Each component of the magnetic field vector is driven by an independent random process to model uncorrelated noise along each sensing axis, while maintaining the same root mean square value.

The stochastic noise η_{total} follows a colored noise structure similar to that adopted for the Sun sensor, consisting of white, pink, and brown noise components. The noise magnitude is derived from the noise spectral density specified in the datasheet ($N_d = 16$ nT/ $\sqrt{\text{Hz}}$). From the sensor sampling period T_s , the standard deviation is evaluated:

$$\sigma = \frac{N_d}{\sqrt{2T_s}} \quad (17)$$

The sensor update rate should be 25 Hz, however, since the overall model operates at a sampling frequency of 10 Hz, the effective magnetometer update rate is reduced accordingly to 10 Hz, and $\sigma = 35.7771$ nT. An example of the resulting noise is in Fig. 7.

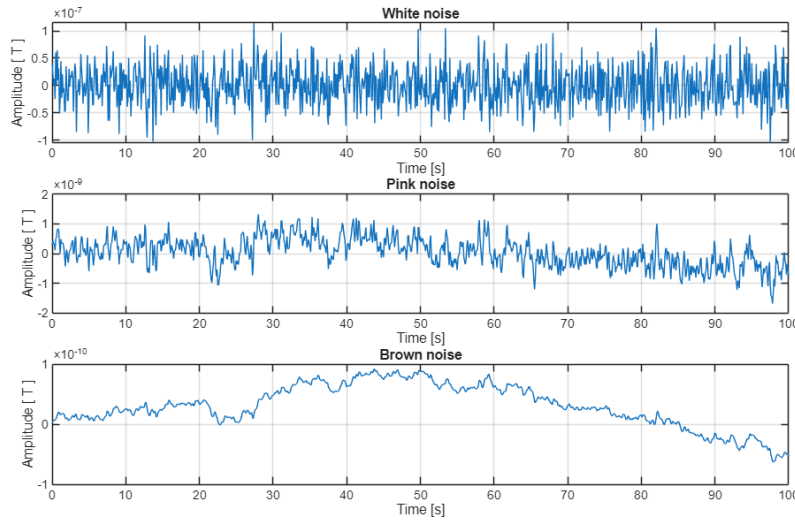


Figure 7: Noise of the magnetometer

Finally, the magnetic field vector is normalized to retrieve its unitary vector.

3.1.4 Gyroscope

The angular velocity of the spacecraft is estimated using a high-performance Micro-Electro-Mechanical System (MEMS) Vibrating Structure Gyroscope (VSG). The selected hardware unit is the Silicon Sensing CRH02-100 [9].

Table 6: CRH02-100 Gyroscope Performance Specifications [9]

Parameter	Value	Unit/Condition
Dynamic Range	± 100	$^{\circ}/s$
Bias Instability	0.12	$^{\circ}/hr$
Angle Random Walk (ARW)	0.017	$^{\circ}/\sqrt{hr}$
Bandwidth (-3 dB)	50	Hz
Scale Factor Error	± 0.15	% of value
Non-linearity	± 0.03	% of Full Scale
Noise Density	0.15	$^{\circ}/s$ rms (at 50Hz)

This specific model selected for this mission profile offers “FOG-grade” (Fiber Optic Gyroscope) bias instability and low noise characteristics in MEMS form factor, which is critical for maintaining tight pointing requirement.

To realistically simulate the attitude propagation errors, the Simulink[®] model implements a comprehensive measurement chain that accounts for deterministic errors and stochastic noise. The measured angular velocity vector is modeled as:

$$\vec{\omega}_B = ZOH \left[\text{sat} \left(\mathbf{S}_{nl}(\vec{\omega}') + \vec{b}_{dyn} + \vec{\eta}_{tot} \right) \right] \quad (18)$$

where the processing chain involves the following stages, according to [8], Ch. 6:

1. **Delay and Frame Misalignment:** $\vec{\omega}_B$ is first subjected to a discrete delay to simulate the sensor’s internal processing latency and readout electronics delay. Subsequently, it is projected onto the non-ideal sensor frame:

$$\vec{\omega}' = \mathbf{M}_{orth} \mathbf{M}_{mis} \left(z^{-1} (\vec{\omega}_B) \right) \quad (19)$$

$\mathbf{M}_{mis}$ represents the mechanical installation misalignment matrix relative to the S/C body frame, and $\mathbf{M}_{orth}$ accounts for the internal sensing axes’ non-orthogonality.

2. **Scale Factor and Non-Linearity:** The MEMS response exhibits higher-order distortions, particularly at high angular rates. The model explicitly accounts for both the linear Scale Factor Error (SFE) and the quadratic non-linearity:

$$\vec{\omega}^U = \vec{\omega}' + \underbrace{K_{SF} \cdot \vec{\omega}'}_{\text{Linear Error}} + \underbrace{K_{nl} \cdot (\vec{\omega}' \odot \vec{\omega}')}_{\text{Quadratic Term}} \quad (20)$$

where K_{SF} is the linear scale factors (1500 ppm) and K_{nl} contains the second-order sensitivity coefficients (300 ppm), where $\odot$ denotes the element-wise product.

3. **Stochastic Error:** The total noise vector $\vec{\eta}_{tot}$ is obtained by summing two independent spectral components. From the sensor’s Allan Variance characterization the following is obtained:

- **WHITE NOISE:** The high-frequency noise component is defined by the ARW coefficient from Tab. 6. In the discrete time domain with step T_s , it is modeled as Gaussian white noise with standard deviation $\sigma_n = ARW/\sqrt{T_s}$.

- **BIAS INSTABILITY:** Modeled using the value provided in Tab. 6. In addition, a feedback term multiplied by $1/\tau$ is included, with $\tau = 100$ s obtained from the minimum of the Allan plot, to model the bias self-correlation.

4. **Saturation and Discretization:** The angular velocity is clipped to the saturation limit and passed through a zero-order hold to model the frequency rate of 10 Hz.

3.2 Actuators

The actuator configuration consists of three reaction wheels, each aligned with one of the spacecraft body axes.

The reaction wheels are modeled using a nonlinear formulation. This block first computes the actuator command $\dot{\vec{h}}$ based on estimated quantities, namely the angular velocity $\vec{\omega}^U$ and the direction cosine matrix $\mathbf{A}_{B/N}^U$. The actuator command is then converted into actual output torque using the true variables provided by *Environment* and *Kinematics*.

The adopted reaction wheel characteristics correspond to the NanoTorque GSW-600, whose main performance parameters are reported in Tab. 7.

Taking into account these physical limitations, torque saturation has been modeled, leading to a bilinear behavior as shown in Fig. 8.

Due to latency blocks in the *Sensors* subsystem, the first control torque is omitted, to avoid using the control action based on non valid sensor values.

Table 7: NanoTorque GSW-600 performance characteristics [2]

Parameter	Value
Max Torque	± 1.5 mNm
Min Torque	± 0.015 mNm
Max Speed at Maximum Torque	± 3500 rpm
Max Speed	± 6000 rpm
Flywheel inertia	$0.3 \text{ kg} \cdot \text{m}^2$

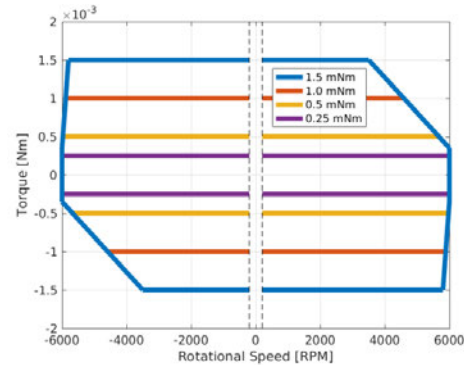


Figure 8: Reaction wheels performance for different torque levels.

4 ADCS Architecture

4.1 ADCS Requirements

Requirement	Value	Rationale
Pointing ($\hat{x}_B$ axis)	1°	Precise tracking
Angular Rate	$0.1^\circ/\text{s}$	Stable tracking

4.2 Attitude Determination

The Attitude Determination subsystem is responsible for estimating the spacecraft attitude in the form of the direction cosine matrix $\mathbf{A}_{B/N}^U$ using sensor measurements. This

matrix is then provided to the attitude control subsystem, which computes the required control torque M_c to orient the spacecraft toward the current target.

The system attitude determination is divided into two operational phases:

- **Sunlit period:** the Sun sensors provide the Sun direction, and together with the magnetometer measures, the spacecraft attitude can be determined.
- **Eclipse period:** the Sun sensors are unable to determine the Sun direction; therefore, the spacecraft attitude is estimated by propagating the last known attitude using the angular velocity measurements provided by the gyroscopes.

The switching logic of the attitude determination algorithm is implemented by checking whether the output of the Sun sensor is $s_B(1) = -2$. This value can only occur when it is explicitly imposed during eclipse conditions, since under nominal operation the Sun sensor output is a unit vector.

4.2.1 q-Method

The selected attitude determination algorithm is the *q-method*, which allows weighting the contribution of the two sensors according to their measurement uncertainties.

The following procedure was adopted to determine the weight α_1 for the sun sensor and $\alpha_2 = 1 - \alpha_1$ for the magnetometer.

Since the output of each sensor is a vector, sensor accuracy can be evaluated by measuring how closely this vector aligns with the true reference direction. Several simulation runs were performed to compute the mean angular error between the true and measured Sun direction, as well as for the magnetic field direction. The results yielded a mean error angle of $\varepsilon_1 = 0.78^\circ$ for the Sun sensor and $\varepsilon_2 = 2.1^\circ$ for the magnetometer.

The sensor weights were determined from the ratio of these mean angular errors:

$$\begin{cases} \alpha_1 + \alpha_2 = 1 \\ \alpha_1 : \alpha_2 = \varepsilon_2 : \varepsilon_1 \end{cases} \implies \begin{cases} \alpha_1 = 0.72 \\ \alpha_2 = 0.28 \end{cases} \quad (21)$$

To validate the effectiveness of this weighting strategy, the attitude estimation performance was evaluated over a time span of $t = 1000$ s, to avoid being affected by the eclipse, using the following metrics:

$$\begin{aligned} \text{sum} &= \sum_t \sum_i \sum_j |\mathbf{A}_{B/N}(i, j, t) - \mathbf{A}_{B/N}^U(i, j, t)| \\ \text{max} &= \max(|\mathbf{A}_{B/N} - \mathbf{A}_{B/N}^U|) \end{aligned} \quad (22)$$

where $|\mathbf{A}_{B/N} - \mathbf{A}_{B/N}^U|$ denotes the matrix whose elements are the absolute differences between the true direction cosine matrix and the estimated one.

This weighting approach led, for example, to the following improvements:

$$\begin{aligned} \text{sum} : & 170(\alpha_1 = 0.5) \rightarrow 168(\alpha_1 = 0.72) \\ \text{max} : & 0.0921(\alpha_1 = 0.5) \rightarrow 0.0920(\alpha_1 = 0.72) \end{aligned} \quad (23)$$

Although the improvement is modest, it was consistently observed across multiple simulation runs, proving a small but steady enhancement in attitude estimation accuracy.

4.2.2 Orbit Propagation

The *Orbit Propagation* block uses the last available estimate of the direction cosine matrix $\mathbf{A}_{B/N}^U$ and propagates it through discrete-time integration of the angular velocity matrix. This procedure follows the same implemented in the *Kinematic* block (see Sec. 2.3).

Initially, in order to avoid adding an additional sensor, the angular velocity was estimated using magnetometer measurements. However, this estimate proved to be highly noisy and significantly different from the true angular velocity vector $\vec{\omega}$, even after applying filtering techniques, as shown in Fig. 9.

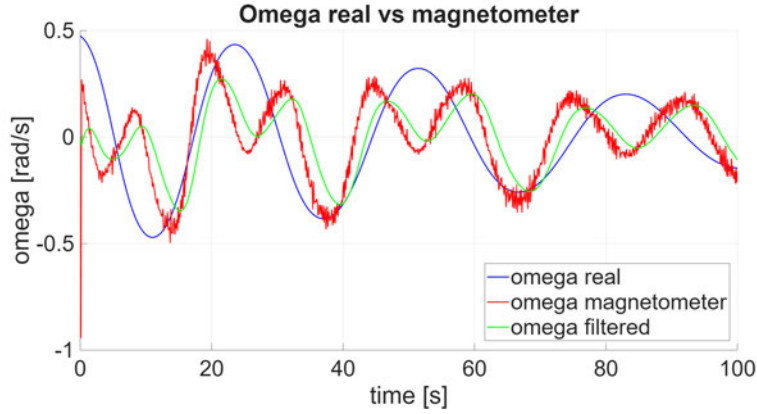


Figure 9: ω_y estimated with magnetometer

In the figure the discrete low pass filter applied is

$$G(z) = \frac{0.5}{\frac{2(z-1)}{Ts(z+1)} + 0.5} \quad (24)$$

With lower cut-off frequencies the amplitude of the signal becomes too low and the time delay increases. With these values of ω the spacecraft randomly moves in eclipse. Designing a filter with the right cutoff frequency was not feasible, since the spacecraft angular oscillation frequencies during eclipse entry cannot be accurately predicted. As a result, the filtered estimate remained too inaccurate to ensure reliable attitude control during eclipse conditions. To maintain active attitude control and reasonable target error, the only viable solution was therefore to include a gyroscope.

4.3 Control Law

The control of the spacecraft is responsible for maintaining the pointing and rotation values bounded and within operational limits. The control receives attitude determination information and commands actuators to exert the correcting torque on the spacecraft.

The mission is divided into 3 separate phases:

1. De-tumbling: slows the spacecraft rotation
2. Re-pointing: moves the spacecraft close to the target attitude
3. Nadir pointing: keeps attitude close to the target

4.3.1 Non Linear control: De-Tumbling and Re-Pointing

The de-tumbling phase involves slowing the spacecraft rotation imparted by the releasing mechanism of the launcher. Release angular velocities can be randomly as high as $30^\circ/\text{s}$; moreover, the initial attitude is randomized, with the target rotated by up to 30° . The

successive re-pointing controller brings the attitude close to the target, while maintaining a limited angular velocity.

Both phases are a non linear controller derived from a Lyapunov function:

$$V = \frac{1}{2} \vec{\omega}^T I \vec{\omega} + k_2 \text{tr}(\mathbf{A}_e^T - \mathbf{A}_e) \quad (25)$$

whose stability is guaranteed by the Lyapunov theorem. Specifically, the de-tumbling is a P-controller on the angular velocity, while the re-pointing is based on the full-state.

$$\vec{u}_{\text{de-tumble}} = -k_1 \vec{\omega} \quad (26)$$

$$\vec{u}_{\text{re-pointing}} = -k_2 \vec{\omega} - \frac{1}{2} k_3 (\mathbf{A}_e^T - \mathbf{A}_e)^V \quad (27)$$

Since stability is guaranteed by design (Lyapunov theorem), the gain of the de-tumbling k_1 and the gains of the re pointing k_2, k_3 have been chosen by tuning them to reduce the actuator force, while maintaining an acceptable error level.

Table 8: Gain in Non-linear control

Gain	k_1	k_2	k_3
Value	0.0005	0.002	0.0002

Switch to Re-Pointing The control manager activates the second controller (re-pointing) once the norm of the measured angular velocity reaches < 0.05 rad/s. This is achieved using moving averages of 10 time steps: the resulting signal is filtered for higher than average oscillation and can be used to determine whether the de-tumbling phase is complete. Also, a shadow mode with a time of 100 s is implemented to avoid too early changes.

Switch to Linear Nadir Pointing The control manager activates the linear nadir pointing when both the angle and the angular velocity are under a certain threshold using moving averages. The implemented conditions are:

- Norm of angular rate below 0.03 rad/s;
- Norm of error below 20° ;
- Shadow mode of 200 s from beginning;
- Previous controller change has happened.

This allows the linear control action to start sufficiently close to its nominal condition.

4.3.2 Linear Control: Nadir Pointing

The main controller is tasked with keeping the attitude close to the target (nadir). The linear control is built using pole placement technique and resembles a PD controller.

Performance	Value	UoM	Rationale
Damping	0.7	-	Reduce overshoot ($< 5\%$)
Settling time	50	s	$< T/2$ to avoid resonance with disturbances

Table 9: Performance of Linear Control

These requirements are transformed into a geometrical boundary in the complex plane. The poles are positioned at the right-most part of the allowed area to limit the action on the actuators and to keep them far from the max torque limitation.

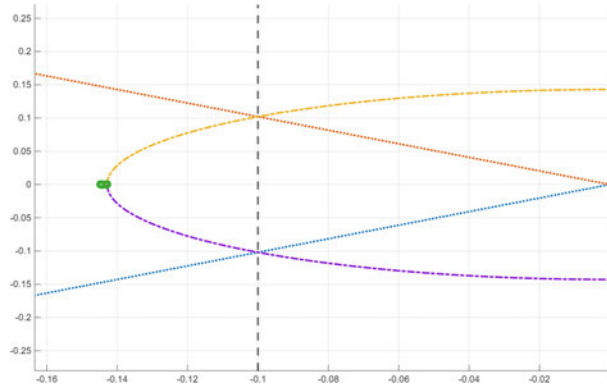


Figure 10: Pole placement technique - Minimal requirements, the green dots are the poles.

The minimum gain for the linear control is:

Table 10: Gain matrix K (up to 4th decimal digit)

$$\begin{bmatrix} -0.1121 & -0.0001 & -0.0000 & -0.0242 & -0.0000 & -0.0000 \\ 0.0001 & -0.0863 & 0.0000 & -0.0000 & -0.0186 & 0.0000 \\ 0.0000 & -0.0000 & -0.0431 & 0.0000 & -0.0000 & -0.0093 \end{bmatrix}$$

Stability Margin and Robustness Using Bode and Nyquist plot to evaluate stability and robustness through the phase margin, it was noted that it was only 7° , well below the standard threshold of 30° .

To achieve a higher margin, the K-matrix is multiplied by 3, increasing the phase margin to about 17° and keeping the control action well below the saturation value. It was not possible to further increase the phase margin given the current actuators.

On the other hand, the system's gain margin is infinite as it does not pass through the negative real axis in the Nyquist plot.

5 Data Analysis & Results

To evaluate the effect of the proposed ADCS system, a statistical analysis has been performed to validate the model and to verify requirements. Below is reported a typical outcome of the simulation.

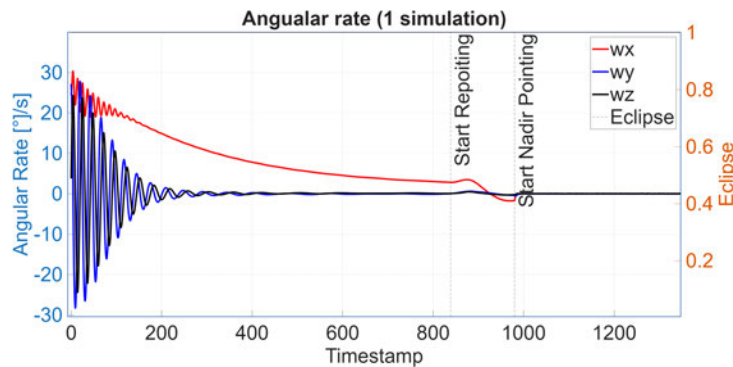


Figure 11: Real $\vec{\omega}$ during non-linear control.

The angular velocity decreases steadily during de-tumbling and re-pointing, reaching values of about $1^\circ/s$.

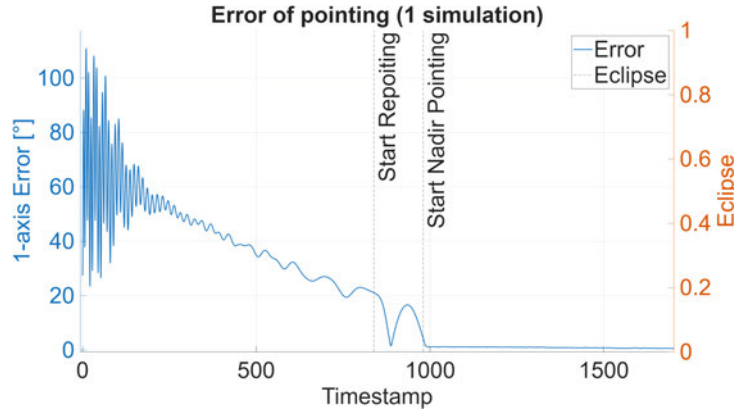
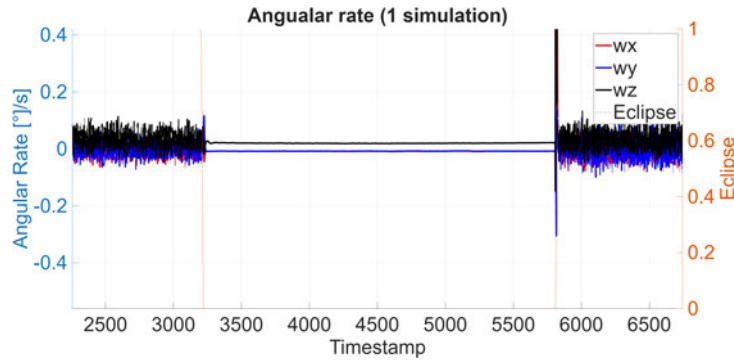


Figure 12: Real attitude error during non-linear control.

The attitude with respect to the $\hat{x}_B$ axis drifts in the de-tumbling phase as the controller acts only on the angular rate. The re-pointing brings the the error to a few degrees from zero.

Figure 13: Real $\vec{\omega}$ during linear control. Focus on both eclipse and Sun-exposed sections.

The angular rate difference between the eclipse (left and right) and the sunlit phase (in the center) is high. In the eclipse phase, it is bounded to about $0.1^\circ/s$ with high oscillation, while in the sunlit phase it is much more constant and close to zero on two axes and with an offset of $< 0.05^\circ$ in the nominal spin axis. During the transition, the torque reaches high value close to the actuators max capacity for a very brief moment.

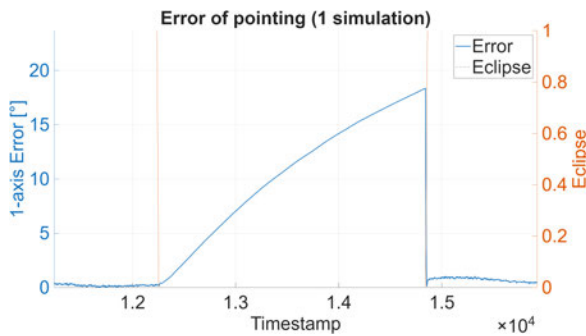


Figure 14: Real attitude error in linear control; focus on the eclipse phase.

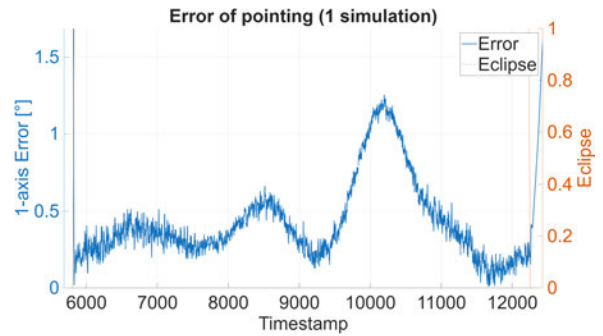


Figure 15: Real attitude error in linear control; focus on the sunlit phase.

The attitude error on $\hat{x}_B$ axis is shown with differences between eclipse and sunlit zones. During the eclipse, the error increases linearly with time, reaching a maximum error of 20° . Due to poor pointing and energy requirement, during eclipse the payload of the mission will not be working. The gyroscope is nevertheless necessary in eclipse to

avoid greater errors. On the other hand, the error during the sunlit phase oscillates but it is almost always $< 1^\circ$. Further analysis will be shown to get actual statistical results of the linear sunlit performance.

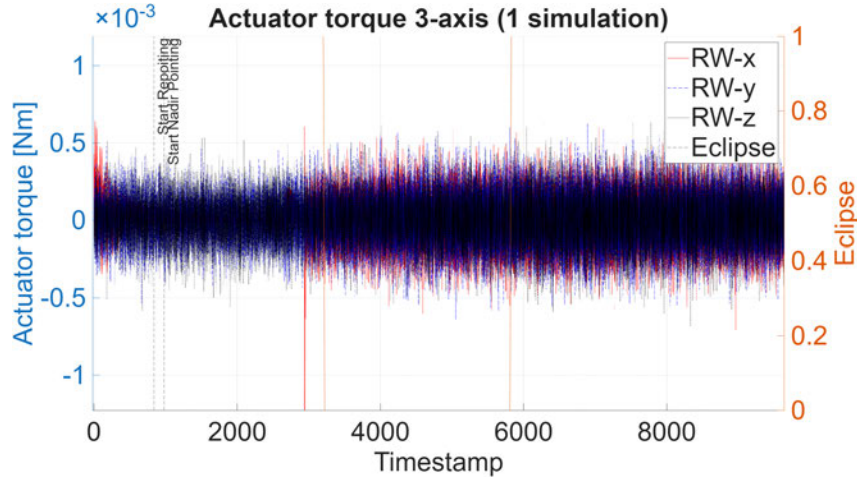


Figure 16: Control torque during 2 period orbit

The control is almost constant in magnitude (changing direction) but its value is well below the saturation of 1.5 mN·m, assuring that the computed control can be produced by the chosen actuators.

The difference with respect to the uncontrolled case is evident. The angular velocity vector continues to oscillate at the initial (high) value and the attitude rotates continuously with enormous angles, with the likely contribution of the unstable gravity gradient.

5.1 Simulations Campaign

A verification campaign comprising of 30 simulations, each with a duration of 2 orbital periods, was conducted to validate the design.

The performance metric adopted is the $\hat{x}_B$ axis attitude error with respect to the nadir pointing direction, called α_x . The dataset is derived from the initial out-of-eclipse phase subsequent to the linear controller activation.

This window was identified as representative (graphically validated by the previous single-run analysis), as the system shows no visible performance deterioration in later sunlight phases. Moreover, considering the significant computational burden of the model, limiting data extraction to the first available out-of-eclipse window was deemed the most efficient approach.

Therefore, the results presented hereafter refer to the first sunlit region following linear control activation. The error profile in this window demonstrates a similar long-term trend across runs, superimposed by stochastic short-term variations and noise.

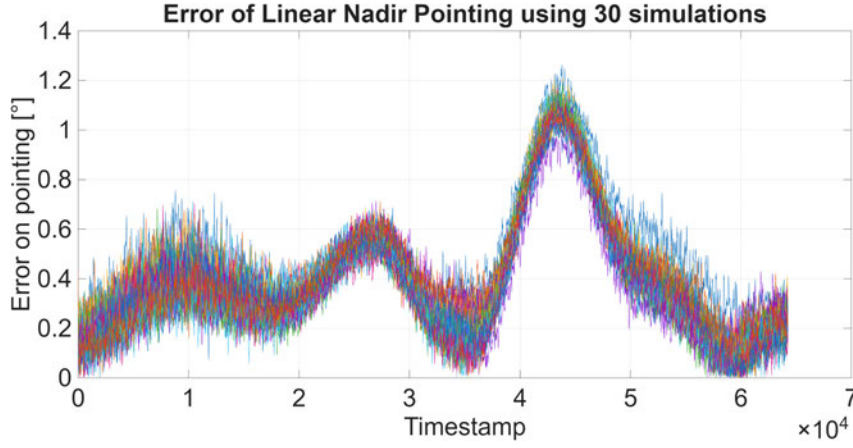


Figure 17: 1-axis Error during Linear control (1st out-of-eclipse period)

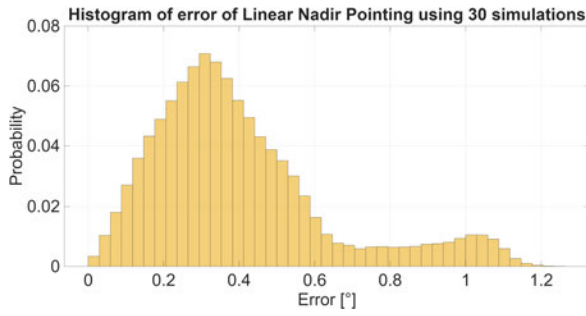


Figure 18: Histogram of Distribution of error (Linear Nadir)

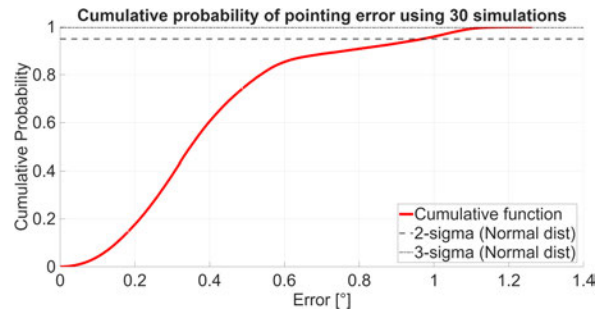


Figure 19: Cumulative Distribution Function of error data (Linear Nadir)

The error is shown to be mostly in the $0^\circ - 0.5^\circ$ region, with some points around 1° . The maximum observed error is 1.25° .

The CDF plot shows that more than 95% of the timestamp has less than a 1° reference error.

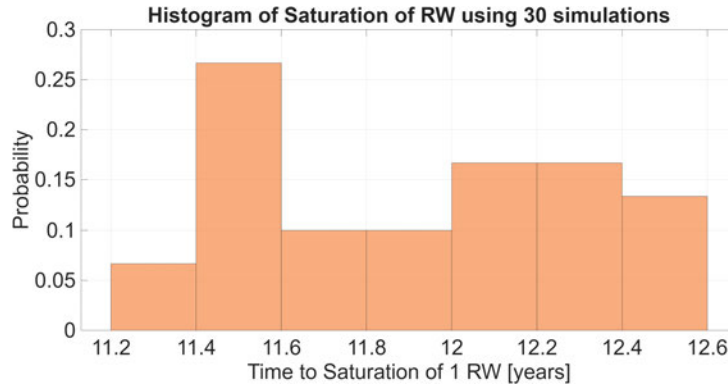


Figure 20: Histogram of Saturation time

The time of saturation of the RWs is computed extrapolating linearly the angular velocity of each reaction wheel. The time of saturation is defined as the time at which the first RW reaches its w^* , where w^* is the value called *Maximum speed at maximum torque* in Tab. 7. The linear model suits adequately the evolution of the angular rate during the linear period (including eclipses). It is shown that the value is over 11 years. It must be noted, that w^* is not the actual saturation speed, just the maximum velocity such that actuators have full capabilities. After that, the torque decreases linearly with

$w - w^*$, making this is a conservative criteria. Nevertheless, the 11 years is longer than the intended life of the spacecraft.

5.2 Statistical Outcome

To show the quality of the control, the chosen parameter is the linear error on the nadir-pointing $\hat{x}$ -axis (α_x). Furthermore, its analyses implicitly contain information about the correct functioning of the previous non-linear stage. In fact, the non-linear block ensures a clean start of the main (linear) controller. In particular, the statistical analysis focuses on:

- Confidence interval of expected value $\mathbb{E}[\alpha_x]$;
- Confidence interval of 95-percentile of α_x , called $P_{95}(\alpha_x)$.

The choice of the simulation number (30) is high enough to allow the use of Normal distribution in statistical analysis of the expected value ($\mathbb{E}[\alpha_x]$) of a generic distribution (like the one of α_x , see Central Limit Theorem). Furthermore, this number allows to filter out the effect of the random initial conditions, noises and uncertainty over some properties (and anything that is modeled with pseudo-random generator).

5.2.1 Pointing Error

Statistic	Confidence Level	Range
CI($\mathbb{E}[\alpha_x]$)	99%	0.74173° – 0.77635°
CI($P_{95}(\alpha_x)$)	95%	0.95617° – 0.99162°

Table 11: Confidence interval for the mean and 95-percentile of the linear error

Both the confidence interval of the mean and the 95-percentile of the error are below the performance requirement of 1°. To confirm that, the accuracy is $< 1^\circ$, a hypothesis test has been performed. It shows that the null-hypothesis $H_0 = \mathbb{E}[\alpha_x] > 1^\circ$ can be rejected with a confidence level of 95%. This is a proof of the effectiveness of the control.

5.2.2 Saturation Time

Statistic	Confidence Level	Range
CI($E[\text{saturationTime}]$)	99%	11.7165 – 12.1126 years

Table 12: Confidence interval for the expected value of Saturation time of the first RW

To confirm that the expected saturation time is > 11 years, a hypothesis test has been performed. It shows that the null-hypothesis $H_0 = \mathbb{E}[\text{saturationTime}] < 11$ years can be rejected with a confidence level of 99%. This is a proof of the adequateness of the actuator, guaranteeing operation-level torque for the lifetime of the spacecraft .

6 Conclusions

From the reported analyses, the presented ADCS architecture is adequate, it guarantees the required pointing accuracy of less than 1° in the 95% of the time stamps, with a Confidence Level of 95%, and to not saturate in the lifespan the reaction wheels. The selected hardware is suitable with the mission requirements.

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